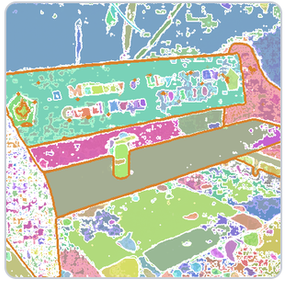
(a) Ask, don't draw: every candidate mask is a token, and the MLLM points at one

Query *"Locate the seat of the bench."*



SAM 3 (frozen)

grid + text prompts



259 candidate masks

point grid SAM 3 text

the MLLM selects

no mask decoder



selected mask

MLLM input sequence Qwen3-VL-4B, 64.2M trainable parameters

one token per candidate mask; bars: its selection score (softmax over all)



(b) One read-out: parts, groups, absent targets

Part

"Locate the fork of the bicycle."



argmax: 1 mask, IoU 0.68

PACO part: +14.5 gIoU

Several targets

"right man and the cake"



σ ≥ 0.90: 2 masks, IoU 0.95

gRefCOCO: +20.8 / +17.2 gIoU

No target

"man on left"



best σ = 0.86 < 0.90: empty

N-acc: 89.6 / 86.0